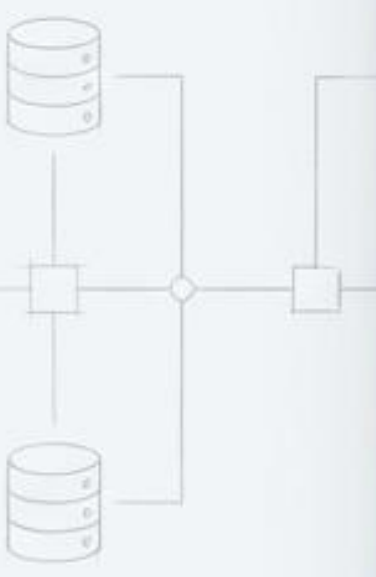




Intelligent Pipeline Orchestration

`harness-ci-cd-customizations v1.0_`

AI-Assisted Development, Automated Governance,
and Enterprise Scaling for JPMorgan Chase.

6 Specialized Agents

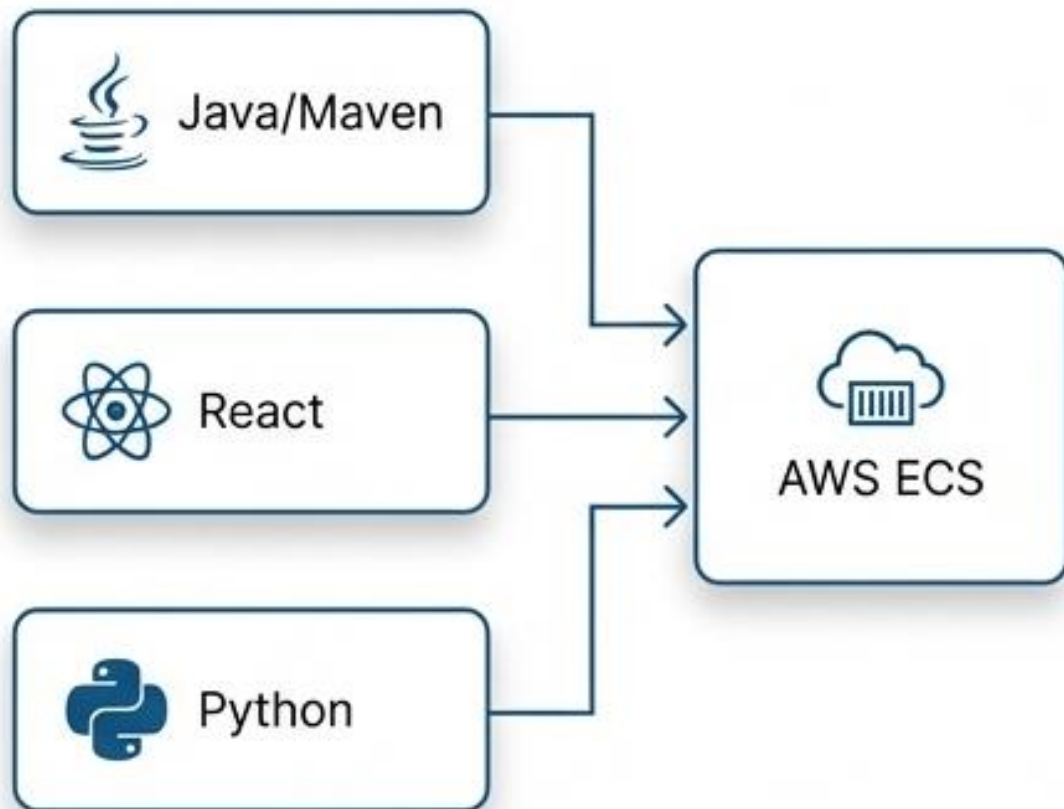
12 Task Prompts

10 Feature Skills

7 Automated Hooks



Velocity & Efficiency




Faster YAML
authoring


Built-in cost
estimation


Automated step
optimization

The Copilot /
Claude
Extension
Engine

6 Specialized Agents
12 Task Prompts
10 Feature Skills
7 Automated Hooks
4 Invisible Instruction Sets

Security & JPMC Governance

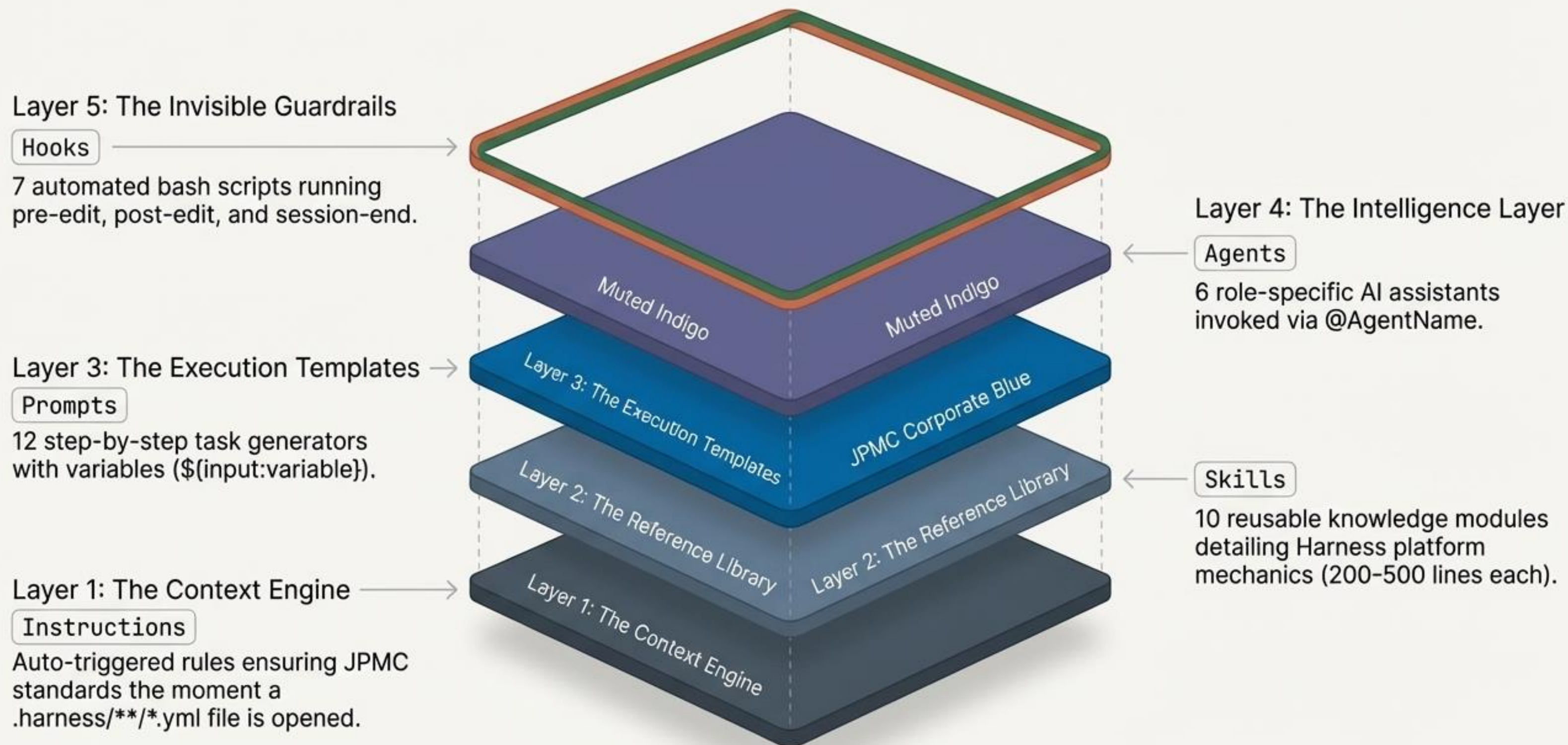



Strict OWASP
scanning


OPA policies
enforced


RBAC
constraints

Macro Architecture: The Blueprint



When to Use What

Tool Type	Complexity	Trigger Method	Best For	Concrete Example
Agents	High (Reasoning/Multi-step)	Manual (@HarnessPipelineDeveloper)	Custom logic, deep debugging.	Review this pipeline for vulnerabilities and fix them.
Prompts	Medium (Structured)	Manual (Use prompt: create-ci-pipeline)	Boilerplate generation, well- defined tasks.	Create a Maven CI pipeline using the standard template.
Skills	Variable (Reference)	Agent lookup / Manual read	Understanding Harness features.	Show me OPA policy examples.
Instructions	Rule-based	Automatic (File edit)	Enforcing JPMC naming and integrations silently.	Automatically applies JPMC ECS deployment standards.

The AI Agent Roster



@HarnessPipelineDeveloper

The Core Builder

Builds CI/CD from scratch, handles matrices, configures test intelligence.



@HarnessSecurityReviewer

The Auditor

Enforces OWASP Top 10, Snyk/Trivy scanning, AWS Secrets Manager validation, OPA compliance.



@HarnessDeploymentStrategist

The Architect

Designs Canary/Blue-Green flows, ALB traffic shifting, Terraform Enterprise (TFE) deployment.



@HarnessPipelineOptimizer

The Speed Tuner

Manages cache intelligence, step parallelization, Docker layer optimization.



@HarnessMonitoringIntegrator

The Sentinel

Injects Dynatrace SG continuous verification and automated metric rollbacks.

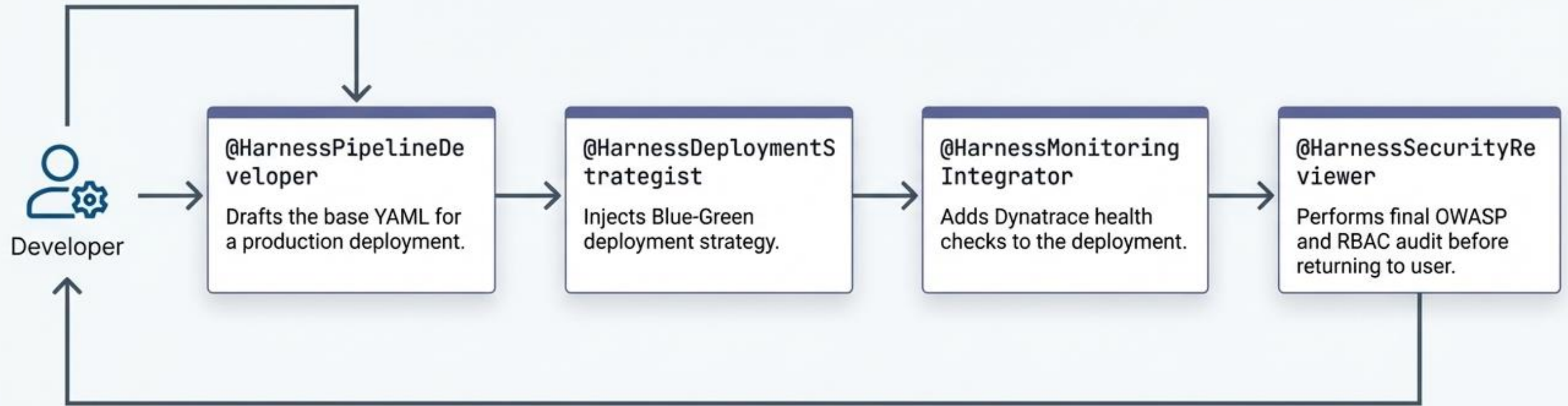


@HarnessTroubleshooter

The Medic

Deciphers log errors, OPA blockages, TFE lockouts, and timeout issues.

Neural Network of Delegation



The Invisible Context Engine

The Input

|

> Use prompt: create-cd-pipeline

`${input:pipelineName}`

`${input:environment}`

`${input:service}`

The Invisible Context

.harness-cd.instructions.md

-  Rule: Enforce ECS via TFE (No direct API).
-  Rule: ServiceNow approval mandatory for 'prod'.
-  Rule: Dynatrace Continuous Verification required.

Prompt Catalog

Creation: create-ci-pipeline

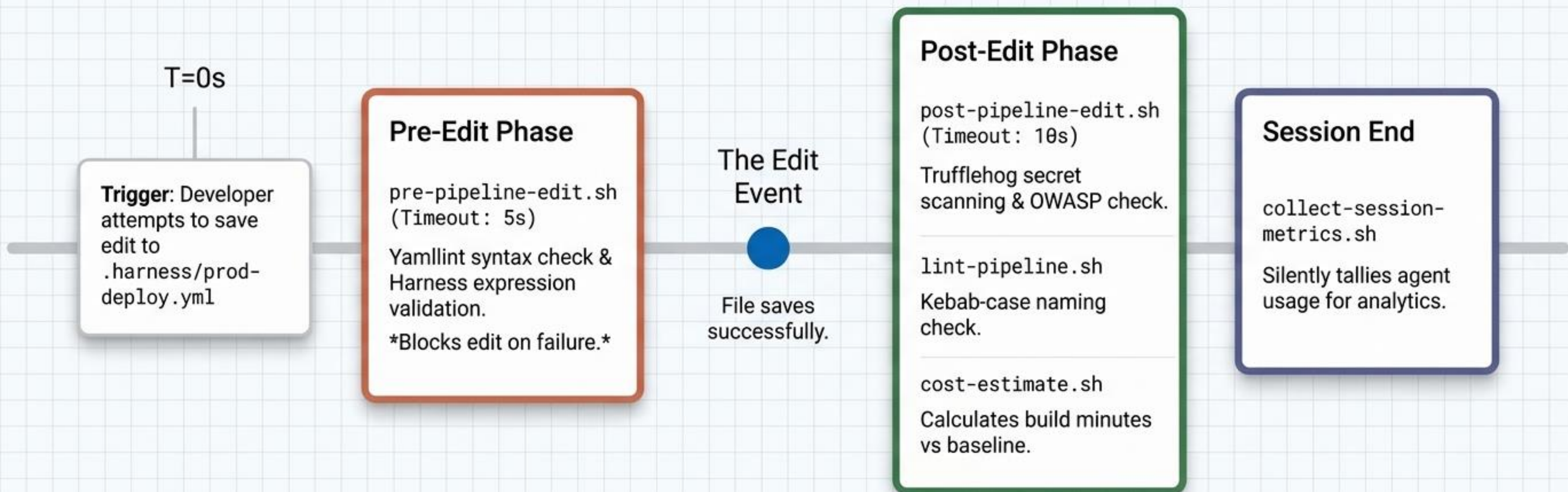
Deployment: add-approval-gate

Security: configure-secrets

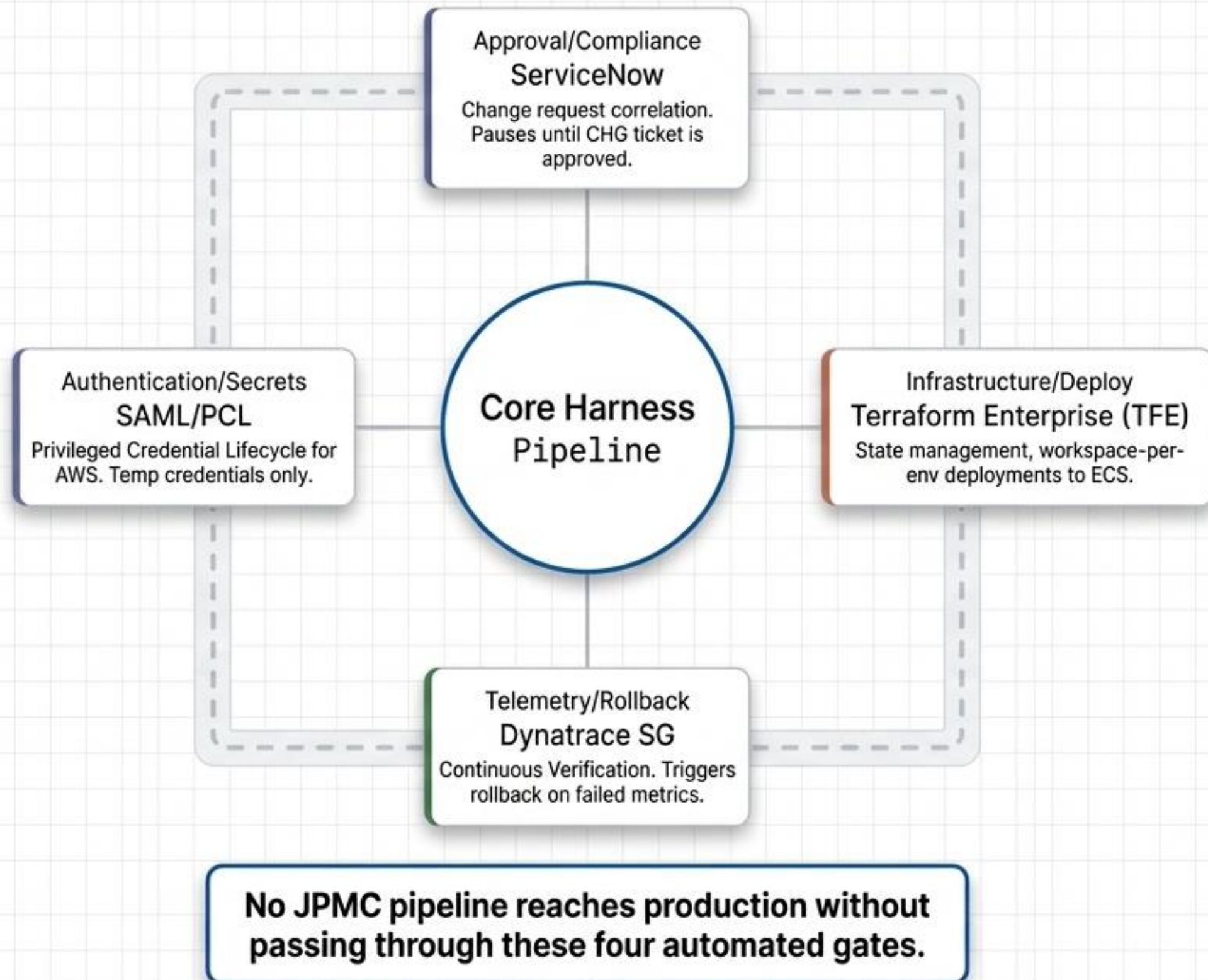
Debugging: optimize-pipeline

The Invisible Guardrails

Event Timeline

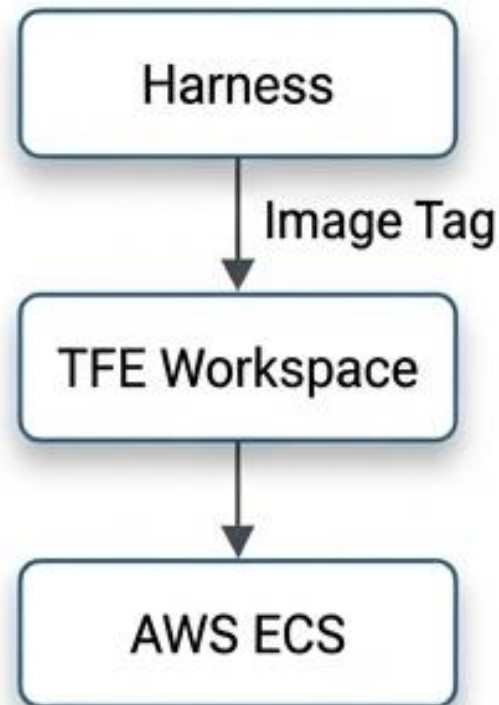


JPMC Enterprise Patterns Architecture



Infrastructure & Security Pillars

Terraform Enterprise (TFE) ECS Deployment

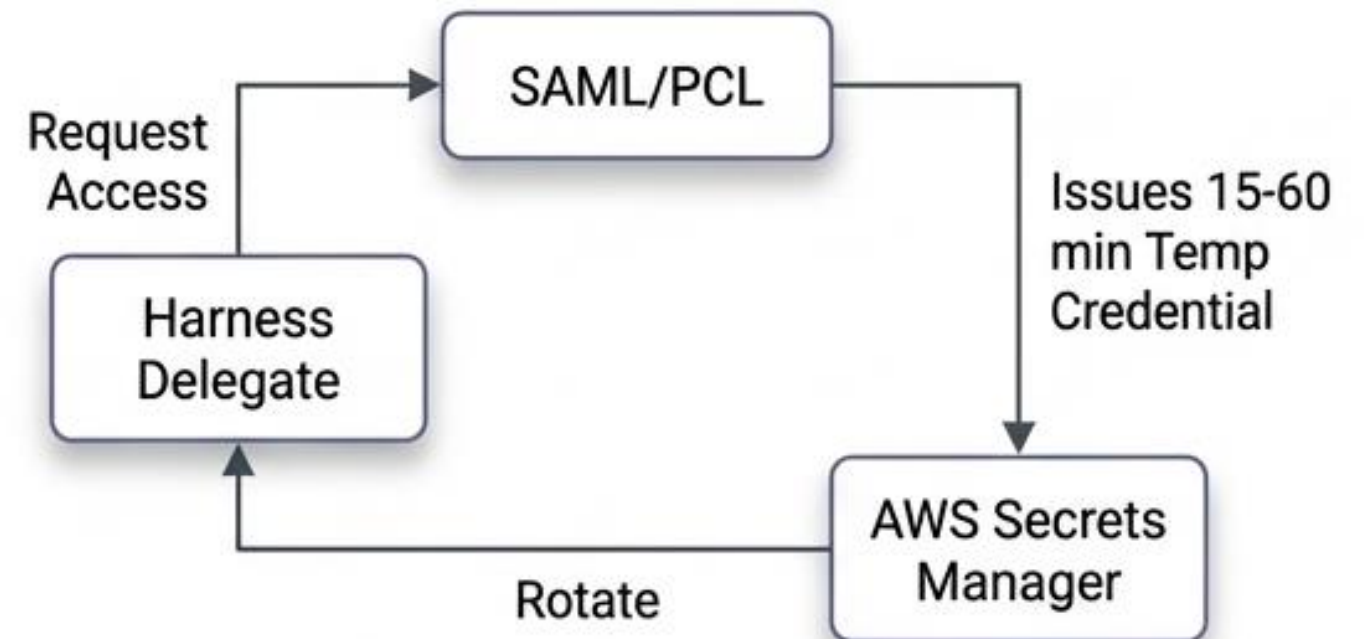


Why: Direct Harness ECS deployments are prohibited to enforce state management and RBAC.

How: Workspace-per-environment pattern. Harness passes the image tag; TFE executes plan -> approval -> apply.

Rule: Rollbacks happen via TFE workspace, not Harness.

SAML/PCL Authentication



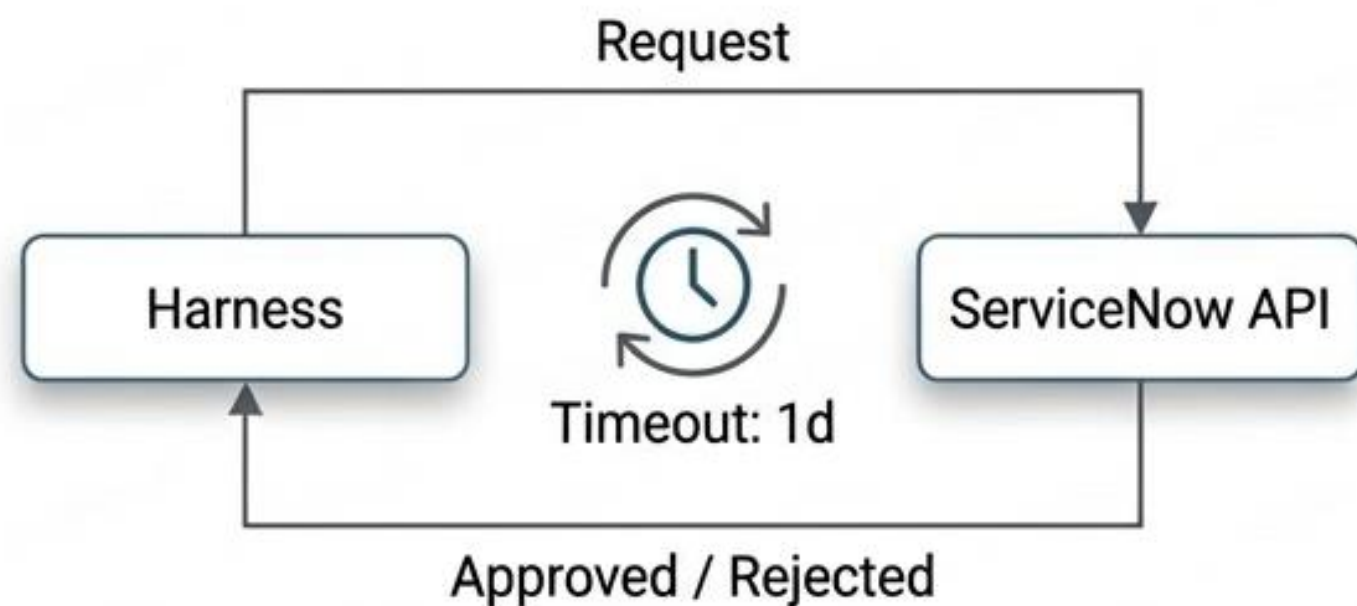
Why: Direct IAM credentials are banned.

How: SAML auth generates temporary AWS credentials. PCL manages automatic rotation.

Rule: All AWS access (Secrets Manager, ECR, CloudWatch) must route through the configured Delegate.

Compliance & Telemetry Pillars

ServiceNow Integration



Requirement: A documented Change Request (CHG) ticket is required for all production deployments.

Mechanism: The approval gate polls the ServiceNow API. It waits up to 1 day for manual organizational approval. Blocks deployment if rejected or timed out.

Dynatrace SG Continuous Verification

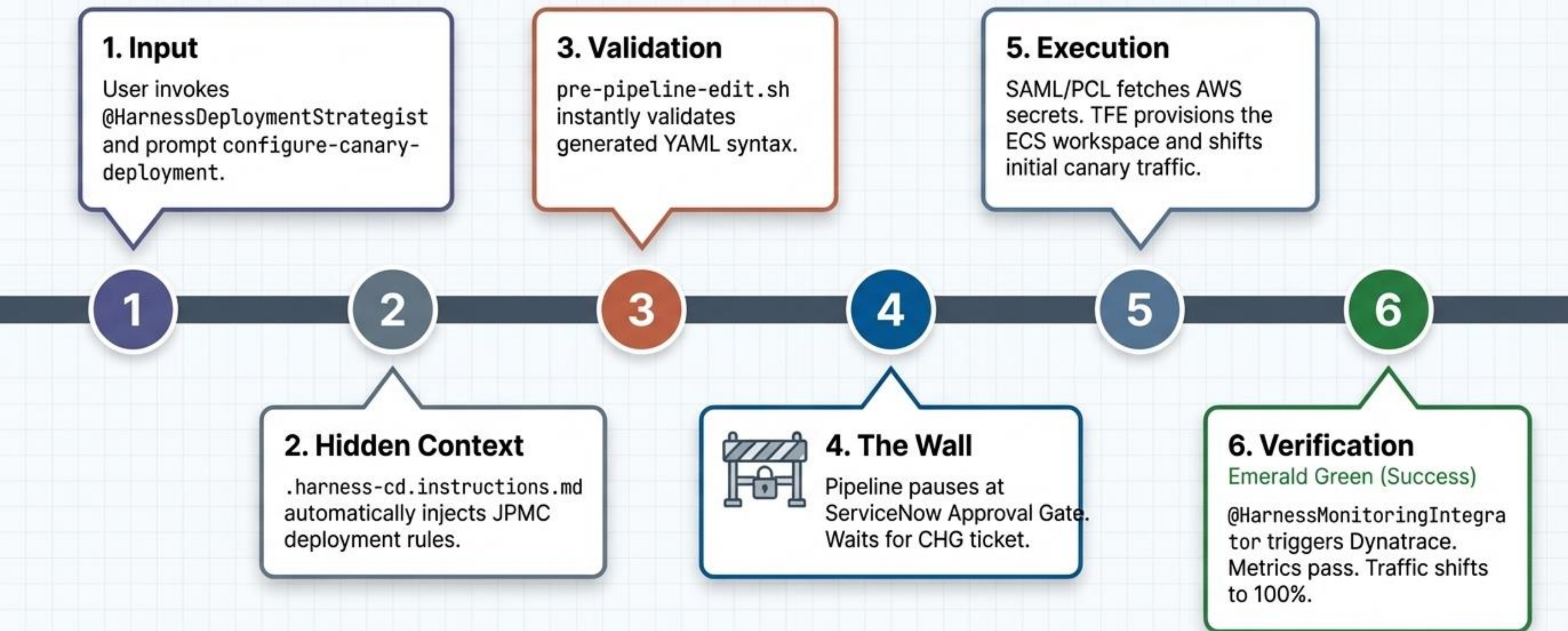


Requirement: Automated verification of application health post-deployment.

Mechanism: Integrates via `.apps.dynatrace.com` (SaaS Gov endpoint).

Safety: MEDIUM sensitivity default. Monitors 5-10 minutes post-deploy. Failure triggers timeout and automated TFE rollback.

Synthesis: Zero to Production



Debugging & Optimization

The Troubleshooter

Inter

Debugging OPA Policy Blockages

⚠ Error Log JetBrains Mono

```
{"status": "failed",  
  "policy": "require-security-scan",  
  "message": "Missing mandatory security step in pipeline."}
```



@HarnessTroubleshooter
JetBrains Mono

✓ Fixed YAML JetBrains Mono

```
stages:  
  - name: Build  
    steps:  
      - name: Security Scan  
        type: Security  
        spec:  
          tool: Snyk
```

Scenario: Build blocked by OPA policy 'require-security-scan'.

Agent Action: @HarnessTroubleshooter detects a generic run step was used instead of the mandatory 'Security' step type with Snyk.

Result: Generates the exact corrected YAML snippet.

The Optimizer

Inter

Speed & Cost Optimization

15m duration

Cost: \$10/day

JetBrains Mono

↓ 70% Faster

4.5m duration

Cost: \$2.75/day

JetBrains Mono

Scenario: A 15-minute pipeline costing \$10/day in compute.

Agent Action: @HarnessPipelineOptimizer analyzes execution. Injects Cache Intelligence, Test Intelligence, and Docker layer caching.

Result: Time reduced from 15m to 4.5m. Compute savings: \$3,650/year per pipeline.

Desk Reference

System Architecture & Cheat Sheet

📁 .github/

📁 agents/ (Count: 6. .agent.md)

📁 skills/ (Count: 10. SKILL.md)

📁 prompts/harness-prompts/ (Count: 12. .prompt.md)

📁 instructions/ (Count: 4. .instructions.md)

📁 hooks/harness-validation/ (Count: 7. .sh)

Invoke Agent:

@AgentName [task]

Use Prompt:

Use prompt: [prompt-name]

Expression Syntax:

<+pipeline.variables.foo> (Never \${...})

AWS Auth:

Always use SAML/PCL via Secrets Manager.

Support:

See AGENTS.md and CLAUDE.md for workflow guidance.